dataframe.R

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```r
# dta frame - rectangular or 2-dimensional array that can contain different data types
# can be created using the function data.frame
?data.frame
```

```
## starting httpd help server ... done
```

```r
### check.rows checks for the consistency of length and names
Age<-sample(x=12:40, size=15);Age
```

```
##  [1] 28 17 15 13 27 18 36 16 22 26 20 30 29 23 21
```

```r
Salary<-sample(x=0:2500, size=15);Salary
```

```
##  [1] 1374 1727 1721 1517 1800  344 1787 1945  750 2011 1137   91 2351   54 1672
```

```r
Weight<-sample(50:140, size = 15);Weight
```

```
##  [1] 135  66 106 140 121  78  54 120  74  61 133  55 119  91  79
```

```r
Religion<-sample(x=c("muslim","christian","others"),
                 size=15, replace=T, prob=c(0.70,0.18,0.12)
                 );Religion
```

```
##  [1] "muslim"    "muslim"    "muslim"    "muslim"    "muslim"    "muslim"
##  [7] "christian" "christian" "muslim"    "muslim"    "muslim"    "muslim"
## [13] "others"    "muslim"    "muslim"
```

```r
my.df<-data.frame(Weight,Age,Salary,Religion);my.df
```

```
##    Weight Age Salary  Religion
## 1     135  28   1374    muslim
## 2      66  17   1727    muslim
## 3     106  15   1721    muslim
## 4     140  13   1517    muslim
## 5     121  27   1800    muslim
## 6      78  18    344    muslim
## 7      54  36   1787 christian
## 8     120  16   1945 christian
```

```
## 9      74  22    750    muslim
## 10     61  26   2011    muslim
## 11    133  20   1137    muslim
## 12     55  30     91    muslim
## 13    119  29   2351    others
## 14     91  23     54    muslim
## 15     79  21   1672    muslim
```

```r
### Referencing in data frame
# a). using row and column numbers
## Name[i,j]
# ith row(s) and jth column(s)
my.df[1,2] # row 1, column 2
```

```
## [1] 28
```

```r
my.df[,1] # all rows of column 1
```

```
##  [1] 135  66 106 140 121  78  54 120  74  61 133  55 119  91  79
```

```r
## b). using row or column numbers
my.df[,"Weight"]# all rows of weight
```

```
##  [1] 135  66 106 140 121  78  54 120  74  61 133  55 119  91  79
```

```r
### Naming dta frames
### row names
rownames(my.df)<-LETTERS[1:15];my.df
```

```
##   Weight Age Salary  Religion
## A    135  28   1374    muslim
## B     66  17   1727    muslim
## C    106  15   1721    muslim
## D    140  13   1517    muslim
## E    121  27   1800    muslim
## F     78  18    344    muslim
## G     54  36   1787 christian
## H    120  16   1945 christian
## I     74  22    750    muslim
## J     61  26   2011    muslim
## K    133  20   1137    muslim
## L     55  30     91    muslim
## M    119  29   2351    others
## N     91  23     54    muslim
## O     79  21   1672    muslim
```

```r
### Column names
colnames(my.df)<-(toupper(c("Weight","Age","Salary","Religion")));my.df
```

```
##   WEIGHT AGE SALARY  RELIGION
## A    135  28   1374    muslim
## B     66  17   1727    muslim
## C    106  15   1721    muslim
## D    140  13   1517    muslim
## E    121  27   1800    muslim
## F     78  18    344    muslim
## G     54  36   1787 christian
## H    120  16   1945 christian
## I     74  22    750    muslim
## J     61  26   2011    muslim
## K    133  20   1137    muslim
## L     55  30     91    muslim
## M    119  29   2351    others
## N     91  23     54    muslim
## O     79  21   1672    muslim
```

```r
# c). for data frame, we can use the dollar sign
my.df$WEIGHT[2:5]
```

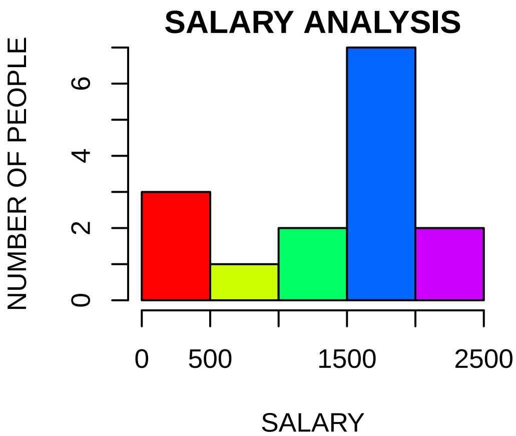
```
## [1]  66 106 140 121
```

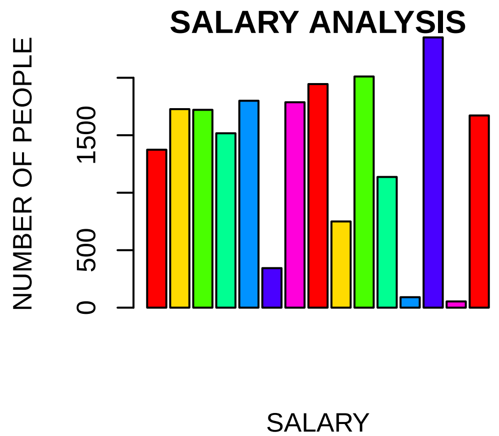
```r
# in my.df,select the weight column and choose from row 2 to 5
my.df$WEIGHT[c(2:5,12:15)]
```

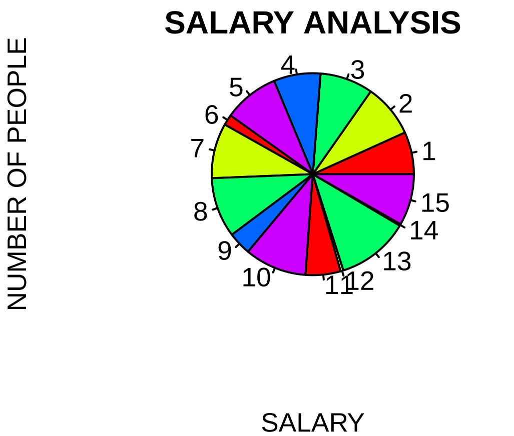
```
## [1]  66 106 140 121  55 119  91  79
```

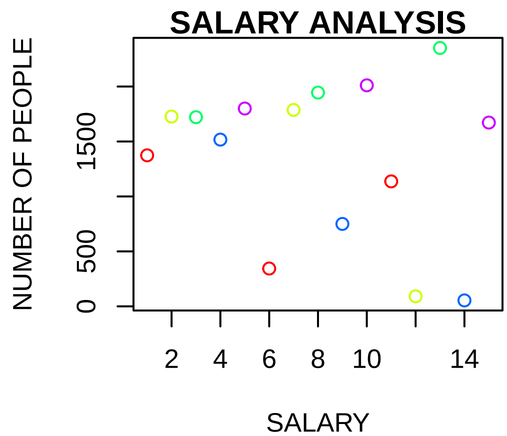
```r
# in my.df,select the weight column and choose from row 2 to 5
# and row 12 to 15


par(mar=c(4,4,1,4),mfrow=c(2,2))
# calling with the $ sign
hist(my.df$SALARY,
     col=rainbow(5),
     xlab="SALARY",
     ylab=toupper("NUMBER OF PEOPLE"),
     main="SALARY ANALYSIS")
barplot(my.df$SALARY,
            col=rainbow(7),
            xlab="SALARY",
            ylab=toupper("NUMBER OF PEOPLE"),
            main="SALARY ANALYSIS")
pie(my.df$SALARY,
     col=rainbow(5),
     xlab="SALARY",
     ylab=toupper("NUMBER OF PEOPLE"),
     main="SALARY ANALYSIS")
plot(my.df$SALARY,
     col=rainbow(5),
     xlab="SALARY",
     ylab=toupper("NUMBER OF PEOPLE"),
     main="SALARY ANALYSIS")
```


SALARY ANALYSIS
NUMBER OF PEOPLE
0
2
4
6
0
500
1500
2500
SALARY


SALARY ANALYSIS
NUMBER OF PEOPLE
0
500
1500
SALARY


SALARY ANALYSIS
NUMBER OF PEOPLE
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
SALARY


SALARY ANALYSIS
NUMBER OF PEOPLE
0
500
1500
2
4
6
8
10
14
SALARY